

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

MANIFOLD LEARNING

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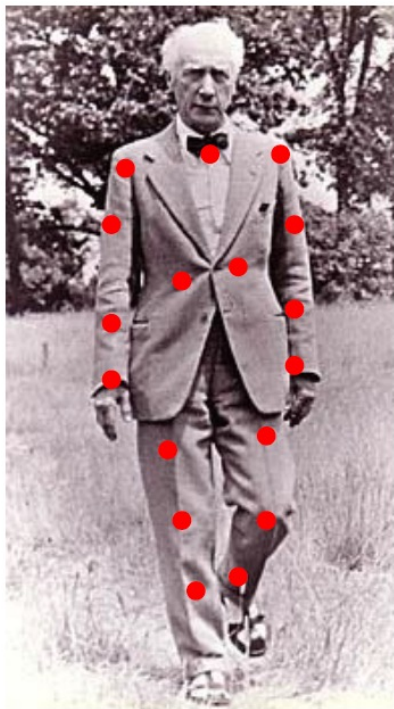
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Low Dimensional Manifolds

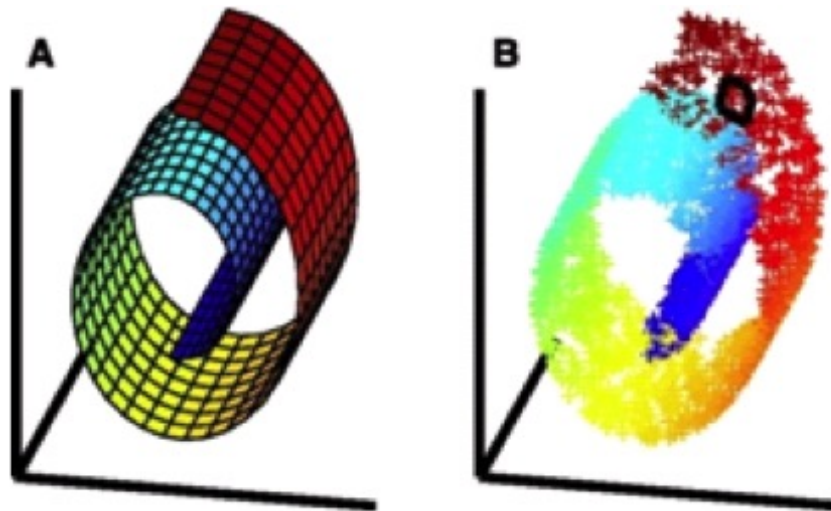
Sometimes data in a high-dimensional space $\mathbb{R}^d$ in fact lies close to a k -dimensional manifold, for $k \ll d$



- 1 Motion capture
 M markers on a human body yields data in $\mathbb{R}^{3M}$
- 2 Speech signals
Representation can be made arbitrarily high dimensional by applying more filters to each window of the time series

This whole area: "Manifold learning"

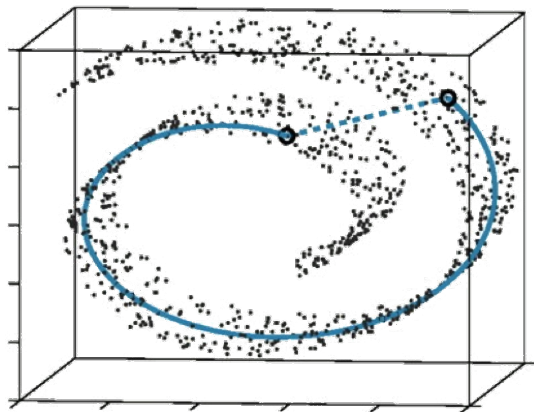
The ISOMAP Algorithm



ISOMAP (Tenenbaum et al, 1999): given data $x^{(1)}, \dots, x^{(n)} \in \mathbb{R}^d$

- 1 Estimate *geodesic* distances between the data points: that is, distances along the manifold.
- 2 Then embed these points into Euclidean space so as to (approximately) match these distances.

Estimating Geodesic Distances



Key idea: for **nearby** points,
geodesic distance $\approx$ Euclidean distance

1 Construct the **neighborhood graph**.

Given data $x^{(1)}, \dots, x^{(n)} \in \mathbb{R}^d$, construct a graph $G = (V, E)$ with

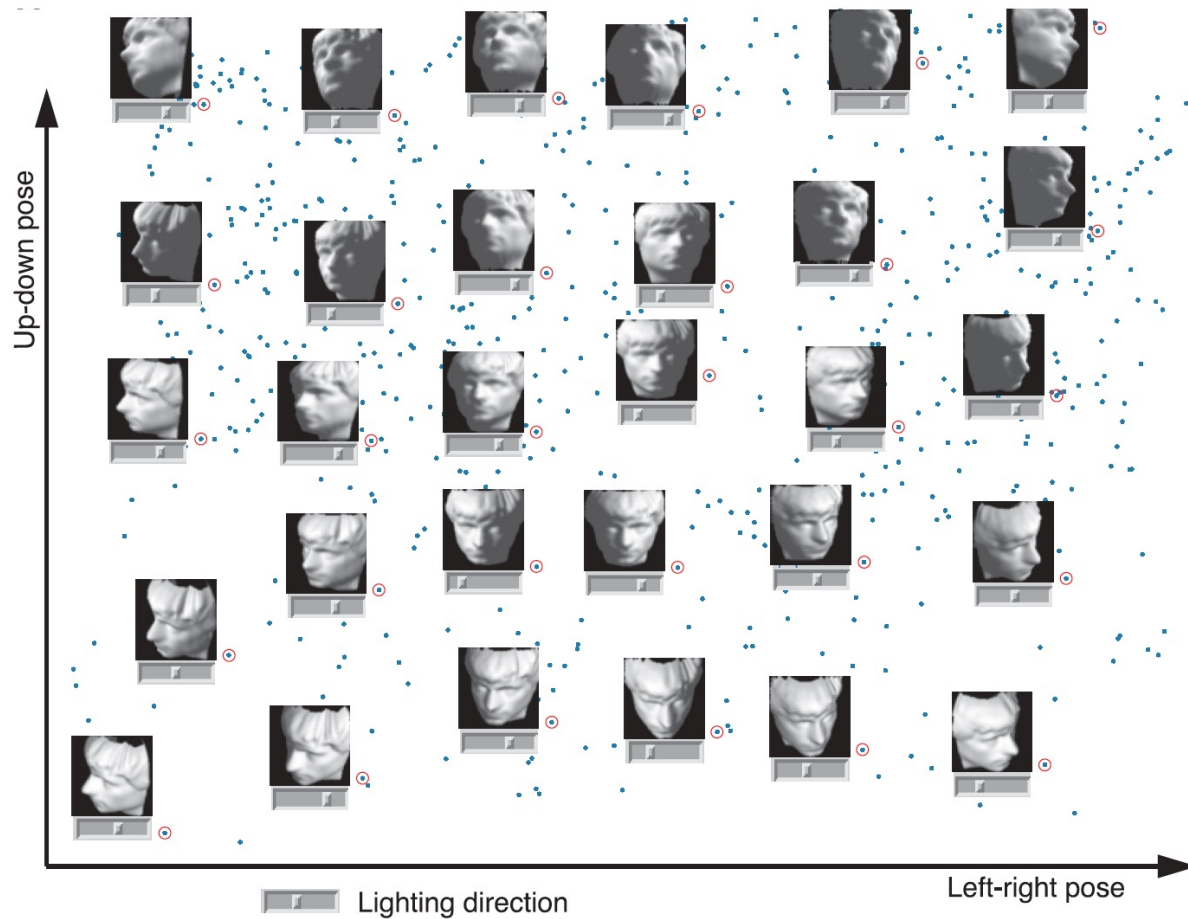
- Nodes $V = \{1, 2, \dots, n\}$ (one per data point)
- Edges $(i, j) \in E$ whenever $x^{(i)}$ and $x^{(j)}$ are close together

2 Compute distances in this graph.

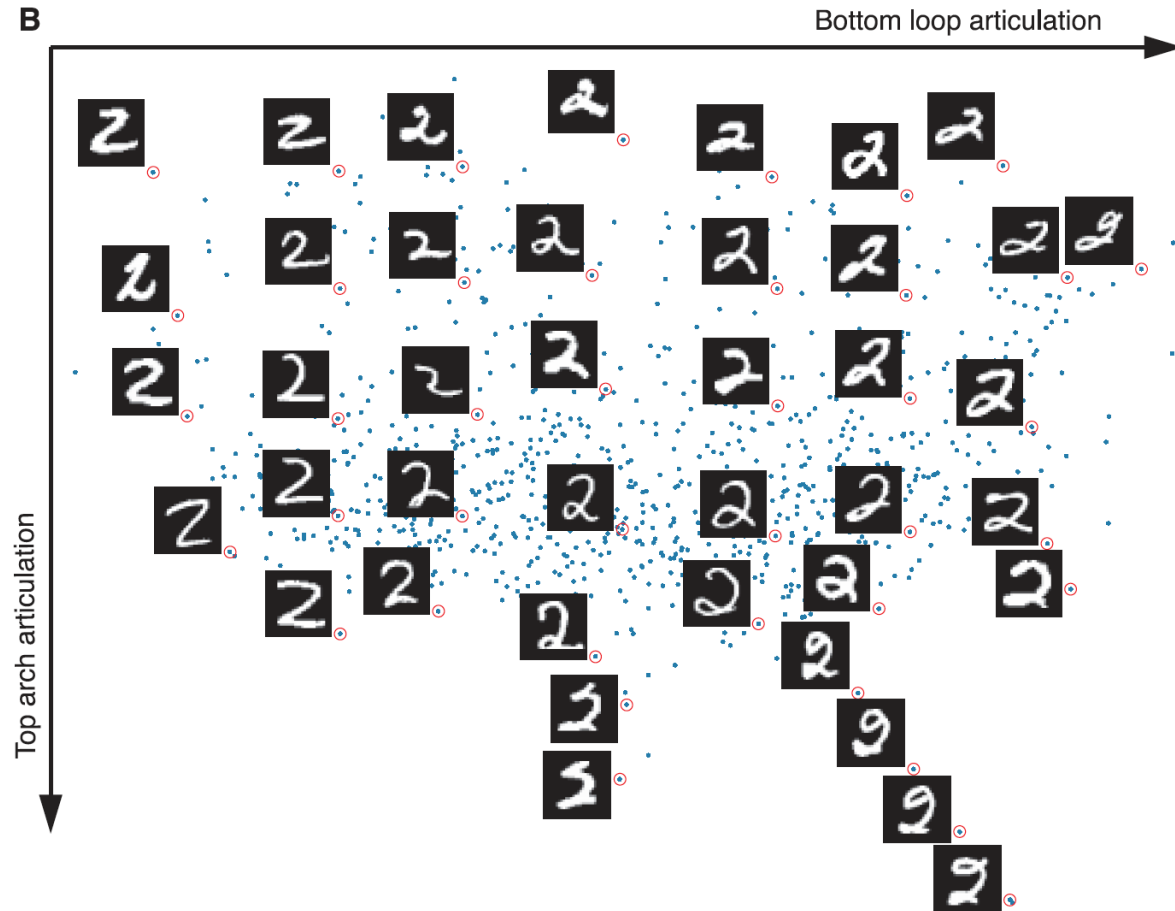
Set length of $(i, j) \in E$ to $\|x^{(i)} - x^{(j)}\|$.

Compute distances between all pairs of nodes.

ISOMAP: examples



ISOMAP: examples



More Manifold Learning

- 1 Other good algorithms, such as
 - Locally linear embedding
 - Laplacian eigenmaps
 - Maximum variance unfolding
- 2 Notions of intrinsic dimensionality
- 3 Algorithms that adapt to data lying on manifolds
- 4 Capturing other kinds of topological structure